

Advance Rolling window Master Roadmap (Pandas)

Goal

Master pandas rolling window from fundamentals to advanced production level usage, including internal behavior, parameters, custom logic, lambda function and ML feature engineering

Phase 1: Rolling window mental model

- What is a window
- Fixed-size window vs time-based window
- Moving context concept
- Rolling vs aggregation
- Rolling vs expanding vs shifting
- Real-world use cases finance ,sensors , healthcare, user behavior, ML Features

Phase 2: Core Rolling API

- Understanding Series.rolling()
- Understanding DataFrame.rolling()
- Rolling object lifecycle
- Syntax patterns
- Default behavior
- Return objects
- How rolling connects with aggregation function

Phase 3: Window parameters Deep dive

- window parameter
- integer windwos
- Offset/time windows
- min_periods
- center
- closed
- right closed
- left closed
- both closed
- neither closed
- step parameter
- axis behavior
- method parameter

Phase 4: Internal Mechanics

- How Pandas build windows
- windows boundaries
- Index-based windows
- Position-based windows
- Time-based window
- Handling irregular timestamps
- Window alignment rules
- Missing values inside windows

Phase 5: Rolling Aggregation Methods

- rolling().mean()
- rolling().sum()
- rolling().min()
- rolling().max()
- rolling().std()
- rolling().var()
- rolling().median()
- rolling().count()
- Mathematical interpretation of each operation

Phase 6: Custom Rolling Logic

- `rolling().apply()`
- Writing custom function
- Function input behavior
- Lambda function with rolling
- Returning scalar values
- Complex calculation inside windows
- Custom statistical measures

Examples.txt - Custom volatility - custom score calculation - anomaly detection logic

Phase 7: Advanced Custom functions

- Passing additional arguments
- Named functions vs lambda
- Reusable rolling function
- Performance consideration
- Avoiding unnecessary Python loops

Phase 8 Time based rolling

- DatetimeIndex requirements
- Rolling by days, hours, minutes
- Difference between `rolling(7)` and `rolling('7D')`
- Calendar-based window
- Irregular event data
- Financial time-series examples

Phase 9 Grouped rolling

- `groupby().rolling()`
- independent windows per group
- Multi-user analytics
- Multi-device monitoring
- Customer behavior features

Phase 10: Rolling feature Engineering for Machine learning

- Lag features
- Moving averages
- Recent activity features
- Rolling statistics
- Trend features
- User-level rolling features
- Preventing data leakage

Phase 11: Advance Window concept

- Rolling correlation
- Rolling covariance
- Rolling regression ideas
- Weighted rolling calculations
- Exponential Weighted windows
- Comparison between rolling and EWM

Phase 12: Performance Engineering

- Vectorized rolling operations
- When `apply()` becomes slow
- Optimized custom function
- Memory considerations
- large dataset strategies

Phase 13: Production patterns

- Building reusable rolling pipelines
- Validation of window features
- Handling missing timestamps
- Testing rolling logic
- Documenting assumption